Abstract

This paper discusses how the Generative Artificial Intelligence (GenAI) can be applied in the academic context and if this technology should be viewed as an opportunity or a threat to education. The results of the study, which is conducted on the basis of the systematic literature review of 21 recent articles, can be summed up as follows: AI applications like ChatGPT can potentially support differentiated education, help the students to produce superior texts, and enhance the efficiency of teachers. Nonetheless, some ethical and pedagogic issues have been noted in the paper including plagiarism, data privacy, bias, and overreliance on AI. According to the article, AI can become a potent educational tool with the help of clear regulations, educators training, and responsible use of AI.

Keywords: Artificial intelligence (AI),Pedagogical tool, Generative AI, Classroom, Data privacy, Responsible AI Use, Personalized Learning.

I. INTRODUCTION

The high rate of Artificial Intelligence (AI) and Generative Artificial Intelligence (GenAI) development has become a significant change within the educational practices nowadays offering the means of teaching, learning, writing academic papers in a new way (Rasul et al., 2023; Pons, 2023). In recent years, AI-powered applications like ChatGPT and intelligent learning systems have gained more and more popularity in the classroom setting with the purpose of making the learning process more accessible, engaging, and customized to individual learning requirements (Goel, 2020; Dempere et al., 2023). The technologies could help students to produce academic materials, structure ideas, solve problems, and get feedback in real time that is done automatically. Therefore, AI has become a potentially disruptive pedagogical technology that will be able to redefine conventional learning approaches and enable more flexible and student-centered learning experiences.

Recent studies also indicate that AI-based learning systems can improve the learning experience of students by supporting inquiry-based teaching methods, immediate feedback, and the creation of interactive learning situations (Xie, 2023; Su and Yang, 2023). The tools can be used to allow students to clear up their doubts immediately, perfect their answers and enhance conceptualization in the process of learning. Moreover, Generative AI can be used to support the students in their academic writing, including grammar, coherence, organization, and clarity of writing work (Getto et al., 2025). This assistance is especially helpful to the learners with problems in language proficiency or academic writing conventions because it enables them to construct systematic and properly structured academic content in a more effective manner.

Instructionally, AI technologies are also very beneficial to the teachers. Research has shown that instructors can use AI-based applications to produce lesson plans, as well as to create quizzes, summaries of learning content, and assessments (Molina et al., 2024; Misra and Li, 2024). AI can save teachers time in the repetitive administrative procedures, including grading and documentation, to spend more time on student-centered support and focused teaching sessions. Moreover, the learning environments provided by AI support students by means of adaptive support mechanisms that offer specific support, depending on the level of performance and learning styles of each student (Abdelghani et al., 2023). These features make AI a potential instructional partner that can be effectively used along with conventional teaching and enhance the overall teaching success.

Besides enhancing the performance of instruction, AI-assisted teaching models are also being considered in the K-12 and higher education settings (Medina, 2025). By integrating technologies of

AI, an educator can analyze the data on the student performance and implement specific interventions, which can improve the learning outcomes and academic performance (Su & Yang, 2023). It is demonstrated that AI-based inquiry teaching methods exert a positive effect on student engagement, problem-solving skills and interaction with academic content (Xie, 2023). Furthermore, AI-based learning tools can also help to make education more accessible and affordable to students with various socio-economic backgrounds and provide them with individualized learning experience outside the classroom setting (Goel, 2020).

In spite of the pedagogical possibilities, AI integration into educational systems, has a number of ethical, cognitive, and institutional issues that need to be tackled to make the implementation responsible. The matter of data privacy and security is one of the main issues that have been pointed out in the literature (Khan, 2024; Miah, 2025). AI-based learning systems can receive and analyze a significant volume of student data to deliver personalized learning experiences. Such information can be the academic performance, behavior patterns, as well as personal information. Certain issues are raised on the manner that this information is stored, shared, and secured against unauthorized access. Poor governance systems and data protection policies might provide privacy risks and misuse of information to the learners.

The other significant ethical issue that comes with the use of AI in education is the problem of algorithmic bias. According to the studies published in the article Artificial Intelligence in Education: Addressing Ethical Challenges in K-12 Settings (Akgun and Greenhow, 2022), AI systems that have been trained with biased data may discriminate against some groups of students and benefit others. It can result in the inequitable learning opportunities and support the existing educational inequalities. Hence, to ensure equitable and inclusive learning environments, it is necessary to address the problems concerning fairness, transparency, and accountability in AI systems.

Academic integrity is another major problem that might arise in relation to AI-assisted learning. This has also challenged the question of plagiarism, authorship, and originality in academic writing due to the ability of the Generative AI tools to produce high-quality written texts (AlAfnan, 2025). Students are able to misuse AI based tools and write complete tasks/responses without thinking independently thereby compromising the validity of their academic assignment. The excessive use of the AI-generated products can result in the reduction of the ability of students to think critically, be creative, and capable of solving the problems independently (Cela et al., 2024; Kanibar et al., 2023). This addiction may facilitate the learning behavior whereby the students are concerned with the quicker solutions and not how learning material is understood and comprehended.

Secondly, it is evident that the successful use of AI devices in the classroom will need teacher preparedness and institutional policies. It has been discovered that teachers can contribute to the education of students regarding the use of AI in appropriate ways and prevent the misuse of such types of technologies (Liu et al., 2025). However, ignorance and absence of training may create resistance or ineffective application of AI tools in learning (Cordero & Cordero-Castillo, 2024). A professional development and training therefore needs to be carried out to equip teachers with the necessary skills to implement AI technologies in the instruction process successfully.

The institutions also have problems of policies and governance frameworks that can be used to control the use of AI in academics. In accordance with Generative AI in the Classroom: From Hype to Reality, it is. Though, currently, some educational institutions lack a properly elaborated set of rules that define what use of AI tools in academic activity is permissible and unacceptable (Pons, 2023), the majority of them do not have them. This ambiguity in policies will create confusion about what is ethical in AI-assisted learning both to students and educators. One can correlate the issue of inequalities, disparities, and technological barriers faced by the learners with the application of AI without the appropriate governance systems (Alfarwan, 2025). This can create barriers to the equitable distribution of the benefits of AI-assisted learning across different groups of socio-economic groups.

In addition to this, research has emphasized the importance of making sure that technological innovation applied in a learning process is balanced with pedagogical effectiveness (Kanibar et al., 2023). Even though the technologies of AI can offer numerous opportunities to improve the teaching and learning process, it is necessary to guide the adoption of AI technologies with the help of the ethical frameworks and institutional policies that will foster responsible usage. To maintain trust and honesty in the system of education, there should be transparency in the process in which AI decision-making is carried out and liability in academic tasks.

Overall, the existing literature indicates that it is possible to use Generative AI to transform pedagogical activities in terms of improved instructional delivery, academic writing aids, and student gains. At the same time, the ethical aspects, the threats to cognition, and the challenges of implementation are to be taken into consideration in order to enhance responsible implementation. In this way, the generalized summary of the previous research is needed to see the pedagogical impact of AI, define the current gaps in literature, and find out whether AI is an opportunity or a threat in the modern classroom environment. This systematic review is aimed at analyzing the proficiency of students in academic writing, identifying the gaps in challenges, developing an appropriate framework to the review procedure, and developing the most significant themes associated with the role played by AI in the classroom. In this manner, the study will focus on determining whether AI is an efficient learning experience or a potential hindrance to the current learning and teaching process.

III. METHODOLOGY

The selected research design of the present paper is the systematic review because the researcher needs to examine the possibilities of the Generative Artificial Intelligence (GenAI) in the classroom and see whether this technology could introduce a pedagogical advantage or a challenge. They were combined through the systematic review approach to include the empirical, review, and policy-based studies on AI-assisted teaching, learning, and academic writing (Alfarwan, 2025; Kanibar et al., 2023). The given plan will enable obtaining a better understanding of the current trends, benefits, ethical concerns, and research gaps in the sphere of GenAI in education (Pons, 2023).

A. Study Selection and Sample description.

All in all, 21 scholarly articles have been analyzed and filtering and selection of all the articles have been done. The inclusion criteria were the articles that included research published in the most reputable academic sources with the extensive scope of learning settings, including K-12 and higher education and teacher education programs (Rasul et al., 2023; Medina, 2025). Moreover, the range of views of different stakeholders that were covered in the chosen articles was rather broad: students, teachers, preservice teachers, and educational scholars (Liu et al., 2025; Misra and Li, 2024).

The inclusion criteria were designed in a way that the articles were applicable and of high quality since they had to be published not earlier than 2020 and 2025 and to be related to pedagogical outcomes, ethical issues, academic writing, or teacher and student perceptions (Dempere et al., 2023; Su and Yang, 2023). The articles that (in the most cases) did not include the application of AI systems to the education sphere were only about technological innovations (Pons, 2023). The selected articles, in that respect, are a compromise measure to both parties of the implementation of AI into the classroom.

B. Data Collection Procedures.

The data gathering was performed by an extremely thorough search of the scholarly databases which include, the Google Scholar, IEEE Xplore, Springer, ScienceDirect, and Frontiers in Education. The keywords were searching the related research as Generative AI in education, AI in classroom, ChatGPT in teaching, AI as a pedagogical tool, and ethical issues of AI in education (Rasul et al., 2023; Khan, 2024). The combination of all the selected articles helped to anchor the most significant facts about the research objectives, practices, demographics of the group of individuals in the research, learning setting, critical findings, benefits of AI use, ethical considerations of the research, and the recommendations of the research (Getto et al., 2025; Molina et al., 2024). To make the studies comparable and consistent, the systematic data extraction model was involved.

C. Research limitations and validity.

The works of different regions, education level, and geographical locations were included to enhance the viability of the findings, and cross-compensation of the perspectives and triangulation of the perspectives became feasible (Kanibar et al., 2023). The fact that peer-reviewed materials were used and the available reviews also increased the reliability of the analysis (Alfarwan, 2025).

The research is however constrained in various ways. The published sources are focused on the literature review and it can ignore the available or unpublished researches. Also, not every study will be carried out with the same type of research design and method of analysis, which may lead to the immediate comparability of results (Pons, 2023). Despite these weaknesses, the systematic review gives a good general impress of the current literature existing and gives some interesting facts regarding the opportunities and challenges of deploying Generative AI into the classroom.

Pedagogical Functions of Gen AI Review.

The selected literature was analyzed in order to identify the existing trends associated with the application of Generative Artificial Intelligence as a pedagogical resource in the classroom (Alfarwan, 2025; Kanibar et al., 2023). The review singled out three key thematic areas, which had a bearing on the adoption of AI in educational institutions: (1) Opportunities to Enhanced Learning and Teaching, (2) Problems and Pedagogical Barriers to Classroom Use, and (3) Ethical, Policy, and Institutional Issues. Each of these dimensions contributes to the answer to the question of whether AI is a powerful aid or a constraint to the learning process (Pons, 2023).

Individualized Learning and Instruction.

The main theme emergence as a result of the systematic review of the selected twenty-one research articles is potential to use Generative Artificial Intelligence (GenAI) to facilitate individual learning and flexible instructional strategies in the classroom environment. The conventional education systems have the tendency to use a uniform model of learning where all the learners are taught the same material irrespective of their unique learning ability, pace, and an academic need. Nonetheless, the empirical research such as Goel (2020) and Xie (2023) indicates that the educational modalities rooted in the AI have a potential to change this traditional paradigm since it provides the opportunities of having personalized learning.

Learning platforms powered by AI can work with real time student performance data, learning trends and engagement. According to these findings, these systems regulate teaching materials, propose the corresponding materials, and offer certain assistance to students who need additional guidelines. Su and Yang (2023) argue that with such adaptive functionality, the students will study at their own pace, therefore, overcoming the constraints of the traditional classroom learning.

The research papers given in Dempere et al. (2023) also confirm that AI-based feedback systems enable students to identify errors immediately and extend their response through the learning process. This is because real time feedback encourages self directed learning and enhances conceptual knowledge since the learners are in a position to think about their mistakes and the knowledge gain.

Besides this, it has been noted that the adaptive learning systems give the students motivation and greater engagement by providing interactive learning which is according to individual preference. According to Alfarwan (2025), AI-based platforms can be utilized to facilitate inclusive learning by considering the inclusion of multiple academic skills and reducing differences in learning outcomes.

However, despite the high-pedagogical potential of personalized learning with the assistance of AI, its effectiveness depends on the quality of its work and the degree to which the teachers are using these technologies in teaching. Individualized learning systems can never go to the point of desired education without proper guidance and supervision.

Instructional Supportiveness and Teaching Efficacy.

The second theme which has been encountered in the reviewed studies is that of Generative AI application in enhancing efficiency of instructions and helping teachers in classroom management and teaching processes. Administering the lessons, structure of examination, grading, and creating of content are usually massive administrative duties that most educators have to carry out. As Molina et al. (2024) note, AI tools can provide automation of some of these routine tasks and, therefore, reduce the instruction planning workload.

Continuing on the point, Medina (2025) highlights the introduction of AI can help educators to concentrate on more effective tasks (grading assignments and developing teaching resources), and less on time-intensive ones to dedicate more time to students, mentorship, and one-on-one instruction.

Empirical evidence that has been presented by Misra and Li (2024) reveals that educators may use AI-based tools to develop quizzes, summaries on course materials, and teaching materials depending on the curriculum objectives. These skills come in useful to improve improved production and to create an effective classroom communication.

Besides, the systems that involve AI can assist the teachers in monitoring the progress of students by providing them with the details related to the learning patterns and the progress of their performance. It is an evidence-based approach that enables teachers to make effective choices and address the issues in the academic sector proactively.

Generally, the literature shows that Generative AI may be integrated as a co-teaching assistant that could be used to enhance the regular teaching processes. Nevertheless, its successful execution depends on how well educators are aware of AI tools and their ability to apply this type of technologies within the pedagogical models.

Interaction and Interactive Learning with Learners.

One of the most important themes, which are present throughout the reviewed literature, is the importance of Generative Artificial Intelligence (GenAI) to improve the engagement of students and make the process of learning in the classroom environment more interactive. The traditional teaching methods are highly reliant on the passive methods of learning whereby students do little more than listen to lectures and follow specific orders. According to recent authors, classrooms can be transformed into student learning experiences with the use of AI-enabled tools, such as ChatGPT and adaptive learning platforms.

A study conducted by Rasul et al. (2023) reveals that the application of AI tools allows students to approach academic material more effectively because it allows them to pose questions, receive responses, and find ideas themselves. This interactive learning process will encourage the students to engage in the learning process rather than to be passive receivers of the information. Equally, according to Misra and Li (2024), students who apply AI tools to participate in guided learning experience higher engagement in activities and assignments in the classroom.

The other insightful argument which is brought out in the literature is the capability of AI systems to provide customized learning. According to Goel (2020), AI-based applications have the ability to learn the performance of students and create the learning content to meet their respective learning pace and the level of understanding and, thus, enhance motivation and engagement. Xie (2023) states that the inquiry-based learning strategies with AI support make students active in problem-solving and critical thinkers because it encourages students to find out what solutions to the problem exist on their own.

The AIs powered learning environments also give real-time feedback as well, which is equally significant in ensuring that the students stay interested and understand better. The research conducted by Dempere et al. (2023) indicates that students will have the benefits of the doubts being resolved and the errors being corrected immediately so that the student can make the answers more precise in the real-time. It is a feedback loop that provides a more interactive and enabling learning process.

Furthermore, according to a study conducted by Abdelghani et al. (2023), under the conditions of AI-enabled simulation and interactive tutoring, the learners will be able to practice the theoretical knowledge in the real-life scenarios, which will provoke the experimentation and the creativity. These tools enlarge the conceptual informative and promote more of a learning process, but as much as AI-assisted interaction possesses a number of advantages, the literature also drawbacks the significance of causing a balance between technological assistance and autonomous learning. The overuse of AI-generated explanations can diminish the desire of students to get to know concepts critically. Thus, teachers should make sure that AI tools are applied as the assistive learning tools, not as the substitutes of active cognitive processes.

In general, the reviewed works always imply that Generative AI can greatly improve student engagement through facilitating personalized learning opportunities, interactive interactions, and instant feedback. AI-based learning surroundings can enhance active engagement and learning results when introduced in a responsible manner under the supervision of teachers.

The AI in Academic Writing.

The other interesting theme that has been identified in the literature reviewed is how Generative Artificial Intelligence (GenAI) could be applied in the case of helping students with academic writing in the classroom. Arguably, academic writing is considered to be one of the most important skills in education and a marker of the knowledge, cognitive skills, and the ability of students to convey the ideas they have in an appropriate manner. However, a large population of students have problems regarding format, clarity as well as following the correct academic writing format. One of the solutions to these issues has been viewed as the introduction of AI-based writing applications.

In the research such as The Role of ChatGPT in Higher Education: Benefits, Challenges, and Future Research Directions, the studies report that AI tools can be used to assist students in generating ideas, structuring the essay, and optimizing the use of language. These tools provide a suggestion on grammar errors, rephrasing of a sentence, and word improvement, which contribute to the overall improvement of academic work. Similarly, in their article, *How to Write With GenAI: A Framework to Use Generative AI to Automate Writing Tasks in Technical Communication (Getto et al., 2025) the authors point out that AI-based writing tools can simplify the writing process because it can be used to remove writer block and to produce a coherent text.

AI writing assistants are particularly beneficial in those students that experience issues with language skills or academic structure. The article Dempere et al. (2023) has written under the title The Impact of ChatGPT on Higher Education explains that AI-based programs may assist students in being more articulate and clear in their work as they receive feedback on how their writing may be improved in real-time. This will help the learners to rectify their work in a more effective manner and they will comprehend the academic writing requirements in a better way.

Another research performing a study titled A Constructive Use of ChatGPT in the Classroom: An Empirical Study (Misra and Li, 2024), also seems to believe that AI tools can also play the role of writing partners that would assist students to organize their ideas in a logical manner and be consistent throughout the course of their work. In addition to grammar and organization, complicated information can also be summarized using AI to make academic text more readable and provide outlines.

Also, the article by Cordero and Cordero-Castillo (2024) suggests the utilization of AI to support collaborative writing tasks, making the tone and formatting the same in case two or more students work on the group assignments. This helps the students and the educators save the burden during the editing process.

However, despite the fact that AI-based writing tools possess a decent list of advantages, the literature also demonstrates the disadvantages of using the tools. According to the article written by Cela et al. (2024), excess use of AI-suggestions may result in the decline of the ability of students to learn how to write independently and gain the skills of critical thinking. Students are able to learn to have the habit of automatic corrections as opposed to learning to construct arguments and ideas by themselves.

In addition, as the article by A. Alfarwan states, there is a possibility that the text created by AI is grammatically correct but lacks depth and originality (Generative AI Use in K-12 Education: A Systematic Review, 2025). This leads to the teacher supervision factor when it comes to ensuring that students utilize AI tools as an aiding tool and not a substitute of intellectual work.

Altogether, the literature review indicates that Generative AI might become helpful in improving the quality of academic writing by assisting with grammar, structure, and clarity. Nonetheless, it can only work well when applied in a responsible manner and guided by teachers in order to make them not to replace the students, but enhance their writing skills.

Effect of Generative AI on Student interaction and active learning.

The other theme that has been identified as salient and came out in the literature analyzed is the use of Generative Artificial Intelligence (GenAI) in engaging students and involving them in the classroom learning processes. The student engagement is one of the key pointers to the learning outcomes, academic performance, and holistic participation in the process of learning. The traditional models of pedagogy are usually the passive models of learning in which the students are provided with the information without being engaged in the process of getting involved into the information. The introduction of AI-based tools, in its turn, provides the opportunities of making the learning settings more interactive and engaging.

Goel (2020) states that AI technologies allow fostering student-centred learning and make educational material more individual to the needs of a person and provide personalised feedback. Such a paradigm of adaptive learning encourages students to play an active role in the learning process since they will be provided with the material at their level and pace of understanding. With the help of AI, students can be engaged in course material through real-time assistance and interactive learning.

Molina et al. (2024) state that AI tools can assist in creative learning as they could assist students in raising ideas, simulating real-life scenarios, and deliberating on different viewpoints of the scholarly problems. These features of engagement motivate increased interaction with the learning material, thereby creating motivation and interest.

Abdelghani et al. (2023) discuss the fact that AI tools may be used to transform the conventional classroom setting into an active learning space. The study suggests that the application of AI as an assistant, but not as a substitute of the teaching process, can make students more active participants of the discussions and group tasks at the classroom. Students will be able to brainstorm, answer questions and reduce their knowledge on complex concepts with the aid of AI-based applications.

According to Xie (2023), AI may be implemented in inquiry-based learning, and AI-intense pedagogical methods can assist students to become more proactive in problem solving and have a greater understanding of academic issues. AI tools will also be beneficial in the formation of the analytical thinking and independent learning behaviours as they help the learners to approach the inquiry process in a systematic manner.

According to Misra and Li (2024), AI can be utilized in collaborative learning through group discussions and enabling the students to work on academic tasks in groups. These tools can suggest, summarize and provide feedback on discussions thus enhancing group work and communications amongst students.

Despite the fact that the advantages of using AI in facilitating active learning exist, the literature also conveys some challenges associated with the application of AI. The Kanibar et al. (2023) state that excessive use of AI-generated responses may decrease the student interest in studying academic content. The students will be tempted to resort to automated solutions as opposed to digging deep and finding ideas on their own.

According to Cela et al. (2024), the ability of students to develop one of their arguments may be limited due to the passive absorption of the information generated by AI. Without the right direction, AI devices will contribute to shallow knowledge and not deep engagement.

Thus, whether AI can be used to enhance the levels of student engagement largely depends on whether it is applied in the learning process. Cordero and Cordero-Castillo (2024) also suggest that educators need to design teaching assignments that will compel students to evaluate the results of AI rather than accept them unconditionally.

In conclusion, the reviewed studies have shown that Generative AI can be used to improve student engagement and promote active learning with the help of interactive and personalized learning opportunities. But to ensure that AI supports active learning and not the promotion of dependency on automatic replies, the teacher control and supervision are also required.

The issue of Ethics in Generative AI Applications in Education.

Among the most significant themes, one can single out the ethical concerns that can be traced in the reviewed literature about the use of Generative Artificial Intelligence (GenAI) in the classroom. Although AI technologies have the above-described pedagogic advantages, their application in the education systems poses important ethical issues that should be addressed to facilitate the ethical use of the tools.

The issue of data privacy is one of the primary ethical questions that are raised in such research as Ethical Challenges of AI in Education: Balancing Innovation and Data Privacy (Khan, 2024) and Ethical Considerations of AI in Education: A Comprehensive Analysis (Miah, 2025). AI-based systems of learning tend to collect and process large amounts of student data in order to deliver students personalized learning experiences. This information usually encompasses scholastic performance, discipline and self-statistics. Storage, sharing and security of such information against unauthorized access have issues. Among the threats to privacy and security in the learning institutions is the fact that student information can be abused.

In addition to the problem of data privacy, an issue of algorithmic bias is also a part of another ethical concern discussed in Artificial Intelligence in Education: Addressing Ethical Challenges in K-12 Settings (Akgun and Greenhow, 2022). The AI systems tend to be trained on large datasets, which might include inherent bias in regards to cultures, social or lingual factors. In its turn, this type of systems may discriminate against certain groups of students unwittingly and favor others, unfavorable learning experiences, and worsen already existing educational inequalities.

Another ethical issue, which also occurs on a considerable scale, is the question of academic integrity in the implementation of AI tools during classes. Based on the study offered in How to Write With GenAI: A Framework to Use Writing Tasks in Technical Communication to Automate Writing Tasks (Getto et al., 2025), students are able to misuse AI tools and not think on their own, writing entire assignments or answers. In the same breath, Technical Report Writing Efficiency Using AI-Powered

Tools: Opportunities, Challenges, and Future Directions (AlAfnan, 2025) highlights the possibility of plagiarism since a student wrote the work generated by AI.

Another problem which complicates the ethical aspect of the academic work with the help of AI is authorship. The students who have widely utilized AI-generated text might be unable to trace the extent of their contribution to the final product. The problem of scholarly work responsibility and ownership is presented by this confusion. As stated in Generative AI in the Classroom: From Hype to Reality? The institutes may struggle to create the boundary between what AI assistance should and should not offer and academic dishonesty (Pons, 2023).

The other ethical concern on application of AI that is raised in the literature is the transparency of decision making. The current number of AI systems is highly thought of as black box systems indicating that users lack a complete understanding of the inside decision making processes of these systems. Such lack of transparency renders teachers and learners incapable to estimate the correctness of the AI-generated proposals. Within the same paper, Generative AI Use in K-12 Education: A Systematic Review (Alfarwan, 2025), it is stated that this form of obscurity may lead individuals to trust AI outputs blindly which in turn may bring about misinformation or even incorrect scholarly work.

Moreover, the area of ethical concerns is extended to the long-term cognitive effects of the use of AI. Such research articles as Risks of AI-Assisted Learning (Cela et al., 2024) presuppose that overusing AI programs might decrease the capacity of students to gain critical thinking skills and acquire expertise of handling issues on their own. The dependency on AI-produced responses without doubting whether they are correct or incorrect may turn the learners into passive consumers who are not actively engaged into the learning process.

In the light of these concerns, other scholars agree with the notion of institutional policies and ethical principles in order to regulate the application of AI in education. In the article by Liu and others (Adopting Generative AI in Future Classrooms: A Study of Preservice Teachers Intentions and Influencing Factors, 2025), much focus is given to teacher training in the progression of responsible AI use in students.

In conclusion, despite the fact that the possibilities of Generative AI in the educational practices are enormous, the ethical side of this technology is to be taken into account. The issues of data privacy, academic integrity, bias, and transparency should be taken to ensure that the application of AI technologies does not negatively affect the integrity of the learning process.

AI Integration in Education Institutional Polices and Governance.

The introduction of Generative Artificial Intelligence (GenAI) in learning spaces has raised the question of the dire necessity to be provided with institutional policies and systems of governance that will assist in responsible usage. Even though the use of AI tools is becoming increasingly popular in classrooms, such as ChatGPT and adaptive learning systems, a significant percentage of learning institutions still do not have explicit policies about the things that can be done and those that are ethically right. Lack of combined models of policies also tends to confuse students and teachers on the use of AI in academic activities.

As revealed in his article Pons (2023) in Generative AI in the Classroom: From Hype to Reality? concludes that the speed at which AI technologies are adopted in the education sector has outpaced the establishment of institutional policies capable of regulating and overseeing the use of AI technologies. This kind of polarisation of technological development and policy adoption is one of the biggest impediments towards upholding academic integrity and the responsible use of AI. Based on this, students are frequently left with a lot of questions concerning the potential of using AI tools in an assignment, academic writing, or research but on the other hand, teachers are confronted with the issue of attempting to balance the use of AI technologies within the pedagogical process without affecting the results of learning.

In the article of Kanibar et al. (2023) that was named Generative Artificial Intelligence in Education: A Review of Benefits, Challenges, and Future Directions, the researcher goes to the extent of indicating that the institutional policies are relevant in defining the limits of acceptability of the academic work supported with AI. Such policies should encompass issues of high concern, including plagiarism, authorship, and transparency in regards to AI-generated material. In order to provide an example, the institutions can mandate students to submit reports on the application of AI tools in the completion of assignments to hold them accountable and original.

Besides safeguarding academic integrity, governance systems ought to encompass data security and privacy consequences. AI-based educational systems tend to gather and analyse data related to learners to give them tailored experiences in learning. Khan (2024) in an article titled Ethical Challenges of AI in Education: Balancing Innovation with Data Privacy highlights the importance of proposing effective data protection solutions to limit how organisations abuse or unlawfully use the data of students. The absence of such protection might trigger the violation of privacy and damage the trust towards educational technologies.

The other critical element of institutional governance is the avenues of professional development that are provided to the educators. In its article, Medina (2025) in AI-Assisted Teaching in Higher Education: Challenges and Opportunities, the author states that the faculty readiness is a determining factor in successful AI use in teaching. In the absence of proper training, educators may have to deal with the challenge of guiding students to be responsible and effective in the use of AI tools, which can be used to support the abuse or resistance of the use of AI.

Also, the differences connected with the equitable access to AI technologies should be taken into account when developing a policy. In the article, Generative AI Use in K-12 Education: A Systematic Review (Alfarwan, 2025), the author claims that the disparity in access to digital resources may create educational disparities in students who have various socio-economic statuses. In that respect, institutions must be fairly accessible that is, all learners must have a proportional access to the advantages of AI-enhanced learning conditions.

The governance frameworks should also enhance openness in the decision making of the AI. The majority of AI systems are complicated and therefore can not be comprehended by their end-users because of the complexity of the algorithms they are composed of. According to the article by Akgun and Greenhow (2022) named Artificial Intelligence in Education: Addressing Ethical Challenges in K-12 Settings, the authors mention that such a requirement as transparency is needed to ensure the credibility and impartially of AI-generated results.

In conclusion, the responsible use of Generative AI in the education sector is an antecedent to the creation and implementation of institutional policies and government structures. Being clear, contributing to the professional development of educators, and taking into account the problem of data privacy and equity, educational institutions can enjoy the merits of AI and minimize the threats that it implies.

V. FINDINGS

The systematic review and the thematic synthesis of the identified 21 studies in the area of the utilization of the Generative Artificial Intelligence (GenAI) in the classroom allow obtaining the following results. Pedagogical efficiency, academic writing support, student involvement, ethical issues, cognitive outcomes and institutional readiness are the most significant issues that were explored.

One of the main aims of implementing GenAI into the classroom setting is the enhancement of the efficiency and personalization of learning. Several articles mentioned that AI-based tools, including ChatGPT and adaptive learning systems can deliver real-time feedback, personalized explanations, and online academic assistance (Rasul et al., 2023; Su and Yang, 2023; Dempere et al., 2023). These tools assist students in making clarification of any questions in real time as well as getting writing and solving problems assistance. It was also established that AI-based inquiry-based teaching techniques do have a positive effect on student learning experience as they allow them to learn actively (Xie, 2023).

Regarding writing in academic institutions, a number of articles demonstrate that the students are more informed about the grammar, clarity, composition, and refined language when using AI-assisted writing (Getto et al., 2025; Misra and Li, 2024). Seemingly, the AI-based writing assistant is highly convenient when the student has low academic writing competencies. On the one hand, however, the positive developments are observable at the surface level, but on the other hand one can have less literature to prove the presence of the substantial improvement in the higher level analytical or critical thinking ability.

Nevertheless, it has been concluded in this review that there are common challenges. Among the most severe issues in this case, academic dishonesty and plagiarism, in its turn, is one of those in which the students consult the material produced by AI when developing their papers (AlAfnan, 2025; Kanibar et al., 2023). There is also active discussion of such ethical concerns as the privacy of data, the bias of algorithms, and the lack of transparency of AI systems (Khan, 2024; Miah, 2025; Akgun and Greenhow, 2022). Using AI in colleges and universities cannot always be a clear policy, and this fact causes difficulties between the teacher and the student (Pons, 2023).

The other important conclusion is the cognitive one. Other scholars state that extreme use of AI tools can diminish the degree of critical thinking and autonomous problem-solving skills in the students (Cela et al., 2024). The passive manner of utilizing the AI-generated responses can be the restricting factor on the unactive interaction with the learning material. On the other side, AI may be a support rather than a replacement of thinking as long as the system is implemented being supervised by the teachers (Abdelghani et al., 2023; Cordero and Cordero-Castillo, 2024).

The motivation of teachers and the institutional authorities also proved to be relevant success factors of successful integration of AI. As it has been found, teachers are aware of the AI possibilities and need to be trained, be instructed in the policies, and oriented to the pedagogical implementation of AI in the classroom (Liu et al., 2025; Medina, 2025).

Altogether, the results reveal that Generative AI in the classroom has the potential to provide quantifiable pedagogical opportunities accompanied by ethical, cognitive, and implementation-related concerns regarding the same.

VI. DISCUSSION

The results can be added to the recent academic agreement that Generative AI in its intelligent use can have a relevant and beneficial effect on classroom learning. Artificial intelligence applications make the process more productive, in this case, they may be used to assist academic writing, and to offer personal learning. Nonetheless, the instances of structural constraints can be found in the literature as well, which makes it difficult to implement in education.

Efficiency in learning is among the most frequently mentioned advantages. AI software help to save time on writing papers, answering questions, and receiving other clarifications (Rasul et al., 2023; Dempere et al., 2023). This kind of effectiveness helps students to devote their attention to concept learning rather than mechanical work. Nevertheless, more profound learning is not assumed in the concept of efficiency like in the case of policy-oriented studies (Pons, 2023). Students overrelying on AI-generated products are able to reduce their cognitive functioning.

One of the conflicts is the conflict between the superficial changes and in-depth learning. The application of AI is very useful in grammar correction, structural organization, and stylistic refinement (Getto et al., 2025). However, it is still low in terms of the capacity to offer critical thinking and innovativeness. The research on the risks of critical thinking (Cela et al., 2024) suggests that the evidence provided by the AI can be presented in an academic form, without being conceptually richer. This fact demonstrates the reason why AI can be considered a support system and not an intellectual one.

The ethical issues expand to the problem of AI implementation in a classroom because of the complications. The issue of plagiarism, the lack of authorship, and responsibility cannot resolve it (AlAfnan, 2025). The problem of concern is the ownership of AI-aided academic work and the extent of AI implementation. In addition, it also happens that the information security of students and algorithmic discrimination is also an issue that should be institutionally controlled (Khan, 2024; Akgun and Greenhow, 2022).

It is based on the teacher preparations. According to the studies, professional development and systematization of the implementation are essential to the success of positive AI integration (Liu et al., 2025; Medina, 2025). In the absence of proper training, teachers may abuse AI or become resistant to it.

Equity and access is another problem. The disparity in the access to the AI technologies can also widen the disparities in education, especially in schools that are less resource-rich (Pons, 2023). This is the reason why AI has to be implemented in a policy-based and integrated way. The literature that remains provides a two-sided reality on the use of artificial intelligence (AI) in the classroom. On the one hand, AI provides the innovative opportunities in the sphere of personal education, essay-writing assistance, and teaching success. On the other hand, it does not have ethical issues, risks of cognitive dependence and institutional issues. It is not the technology but its means of application that is pedagogically regulated and controlled.

The work that lies ahead to be done in the area of generative AI ought to be based on the moral principles, professionalism of the teachers, correct policy provisions, and AI literacy among students to ensure that the area is more of an opportunity than a threat. AI introduction can only be positive to human teaching and learning so long as it is offered in a positive manner as opposed to negative.

VII.CONCLUSION

The present paper has discussed Generative Artificial Intelligence (GenAI) application in the classroom and how it affects the performance of learning, academic writing, and teaching methods. The results show that AI applications, including chatGPT and other AI-based learning systems, can be used to facilitate an individual learning process, stimulate engagement among the students, and make the learning process more effective. The AI can also be applied to help students write academic papers, enhance their grammar, clarity and structure, and instructors in planning their lessons and creating their tests.

Regardless of these benefits, the research arrived at the conclusion that there are a number of challenges that are associated with the use of AI in the learning process. The issue of academic dishonesty, copyright of information, data confidentiality, and algorithmic discrimination continue to be critical drawbacks to responsible implementation. In addition, overuse of AI-tools can interfere with students capacity to think critically and problem solving. It is these constraints that lead to human control, which is required to make sure that AI will not substitute the learning process but rather enhance it.

The future of the AI in education is relatively promising especially considering the development of adaptive learning systems that will help in supporting the needs of the diverse students. Integrative effort however needs appropriate institutional policies, codes of ethics and good teacher training. These need to be addressed so that AI will not undermine academic practice and academic outcomes.

On the whole, despite the vast potential of generative AI in enhancing classroom learning, its outcome is determined by the responsible usage, careful application, and regulation over a human in the long-term. When properly used, AI might be a concept of the joint learning technology that would supplement the traditional approaches to teaching and will result in the students evolving.